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A data resource of peptide–protein interaction structures and binding affinities for peptide drug discovery

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Abstract

Protein–protein interactions (PPIs) are important therapeutic targets but remain difficult to modulate with conventional small molecules. Peptide ligands can engage extended interfaces with high affinity and specificity, and advances in AI-enabled peptide drug design have increased the demand for curated peptide–protein interaction datasets. Here we present PPIKB, a knowledge-based data resource of peptide ligand–protein interaction, including structures and quantitative interaction measurements such as Kd, Ki or IC50. These data were manually extracted from 3,894 peer-reviewed articles, 553 public patents, and Protein Data Bank (PDB) records. The quantitative interaction module comprises 21,845 binding measurements for 2,151 protein targets, with standardized identifiers and units, peptide sequence and modification annotations, physicochemical descriptors, assay metadata, and bibliographic links. The structural module integrates 18,005 experimentally resolved protein–peptide complexes annotated with chain assignments, experimental method, resolution, and residue-level noncovalent contacts. The modules are cross-linked via shared UniProt and PDB identifiers to support retrieval of matched structure–measurement records and downstream reuse, including model training and benchmarking, similarity-based retrieval, and comparative analyses of peptide–protein interactions.

Background & Summary

Protein–protein interactions underpin virtually all major biological processes, including signal transduction, intercellular communication, immune surveillance and metabolic regulation. Their dysregulation frequently drives disease pathogenesis, which makes PPIs highly attractive but often technically challenging therapeutic targets[1,2]. Traditional small molecule drugs offer clear advantages such as low production cost, oral bioavailability and favourable membrane permeability, yet their limited surface area and conformational flexibility often preclude effective modulation of the large and relatively planar interfaces that characterise many PPIs[3–5]. By contrast, peptide therapeutics combine structural versatility, high binding specificity and generally low immunogenicity with moderate manufacturing costs, which makes them powerful modulators of PPIs across a broad range of targets[6–9]. Beyond their role as therapeutics, peptides serve as selective biorecognition elements in diagnostic platforms, and their biocompatibility, chemical tunability and stability have driven the development of

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peptide based electrochemical biosensors that offer high sensitivity, rapid response and cost effectiveness for point of care applications[10–13]. Peptides also function as delivery ligands and conjugation scaffolds, enabling targeted delivery and pharmacokinetic optimization through peptide drug conjugates and related formats[14,15]. In addition, peptide functionalization can be used to enhance extravasation and tissue targeting in delivery systems, providing a complementary route to improve in vivo performance of PPI directed modalities[16].

Recent advances in artificial intelligence have reshaped many aspects of drug discovery, particularly through breakthroughs in protein structure prediction and generative modelling. Universal frameworks such as AlphaFold, RoseTTAFold and ESMfold have transformed our ability to model protein conformations, and specialised tools now refine these predictions for more complex systems[17–19]. In our own work, HighFold improves cyclic peptide and peptide protein complex predictions[20], while AFsample addresses peptide flexibility by generating multiple conformational isomers[21]. In de novo peptide design, diffusion based generative models such as RFdiffusion and sequence optimisation platforms such as PepINVENT explore large regions of chemical space, including noncanonical residues[22,23]. Our design platform CycleDesigner, which builds on CycRFdiffusion and HighFold, enables rational design of cyclic peptide binders with experimental validation, and our reinforcement learning framework HighPlay uses HighFold structure predictions to design cyclic peptides de novo[24,25]. Transformer based predictors such as TPepPro further combine sequence and structural features to enhance interaction prediction accuracy[26].

Despite these methodological advances, available data on protein–peptide interactions remain fragmented and uneven in coverage. A number of peptide focused resources, including DRAMP and other antimicrobial peptide collections such as APD, CAMP, DBAASP, LAMP and dbAMP, have assembled large and well annotated sets of bioactive peptide sequences together with functional activities, mechanisms of action and physicochemical descriptors, but they generally emphasise peptide function against microbial strains or cell lines rather than quantitative binding affinities to defined protein targets[27–31]. Complementary resources that focus on binding measurements, such as PDBbind and BindingDB, offer detailed binding data but concentrate primarily on small molecule ligands; peptide ligands are relatively underrepresented and are not systematically prioritised as a distinct class[32,33]. Structural repositories such as Propedia and PepBDB compile extensive collections of peptide protein complexes, yet typically lack the corresponding affinity measurements needed to link structure with binding strength[34,35]. As a result, it remains difficult to obtain a broad and internally consistent view of protein–peptide binding that spans different protein families, assay formats and peptide chemotypes.

Here, we introduce PPIKB (https://ppikb.duanlab.ac/), a large-scale data resource for peptide ligand–protein interactions built through large-scale curation of the biomedical literature and public patents with expert validation. PPIKB integrates tightly cross-linked modules. The quantitative interaction module is a curated collection of experimentally determined binding measurements between peptide ligands and protein targets, linked to standardized protein and peptide identifiers, peptide sequence/modification annotations, computed physicochemical descriptors, assay metadata, and bibliographic information. The structural module assembles experimentally resolved protein–peptide

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complexes from the Protein Data Bank and augments them with residue-level contact and interaction profiles. By enabling retrieval of matched structure–measurement records, PPIKB supports downstream reuse in peptide drug discovery, including data-driven peptide design, model training and benchmarking, and systematic characterization of peptide–target interaction patterns.

Comparison with existing peptide data resources

To contextualise its scale, we compared PPIKB with representative resources widely used in peptide or binding studies (Table 1). Most directly related to PPIKB is the structure-based protein–peptide affinity set of Wang et al. (PpISADS), which manually curated over 350 protein–peptide complexes carrying both an experimentally determined structure and an affinity value for computational peptidology[47]. PPIKB extends this structure-plus-affinity concept by roughly two orders of magnitude (21,845 quantitative measurements and 18,005 complexes) while adding standardized identifiers, peptide modification and cyclization annotation, physicochemical descriptors, assay metadata and patent-derived records. Among structure-only peptide resources, PeptiDB (103 high-resolution complexes) and PixelDB (1,966 non-redundant complexes annotated for binding-mode conservation), together with PepX, PepBind, PepBDB and Propedia, provide curated complex sets but do not link the corresponding binding affinities[34,35,43–46]. General protein–ligand affinity resources such as PDBbind and Binding MOAD include affinity data tied to structures but are dominated by small-molecule ligands, with peptides not curated in a peptide-centric, sequence-based manner[32,48]; and the protein–protein affinity benchmark of Kastritis et al. (144 complexes with measured Kd) addresses protein–protein rather than peptide ligands[49]. PPIKB is therefore distinguished as the largest peptide-specific resource that couples quantitative binding measurements with structural evidence, curated from both literature and patents and cross-linked through shared UniProt and PDB identifiers.

Table 1. Comparison of PPIKB with representative peptide structural resources and general binding databases.

Database	Affinity entries	Structure (non-redundant)	Year
PepX	/	1,431	2009
PeptiDB	/	103	2010
PepBind	/	3,100	2013
PixelDB	/	1,966	2018
PepBDB	/	13,299	2020
Propedia	/	15,126	2023
Wang et al. (PpISADS)	~350	~350	2024
PDBbind	23,010*	23,010	2025
BindingDB	3,051,069*	/	2025
PPIKB	21,845	18,005	2025

** PDBbind and BindingDB primarily focus on protein–small-molecule binding; peptide ligands appear only sporadically and are not systematically curated with peptide-centric annotations.*

Methods

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Data sources and selection criteria

To construct PPIKB in a systematic manner, we implemented a multistage data collection and curation workflow. We first searched major scientific literature databases, including PubMed, Google Scholar and Scopus, and retrieved patent information from Lens.org, Google Patents and the patent offices of major jurisdictions (United States, China, Japan and the European Union). The search strategy focused on peptide ligand–protein interactions and binding affinity, using combinations of terms such as "protein–peptide interaction", "affinity", "surface plasmon resonance", "biolayer interferometry", "fluorescence polarization", "enzyme activity assay" and "microscale thermophoresis", together with quantitative descriptors "Kd", "Ki" and "IC50". Candidate records were manually screened at the title, abstract and full-text levels to identify studies reporting quantitative binding between peptide ligands and well-defined protein targets.

Entries were incorporated only when several conditions were simultaneously satisfied. The peptide sequence had to be explicitly reported and contain no more than 100 amino acid residues. Reported affinity values were required to be numerical and derived from wet-lab measurements rather than qualitative activity readouts. The protein target needed a clearly defined sequence, either directly provided in the source or unambiguously inferred from the reported gene and species. The source of the affinity data had to be a peer-reviewed publication or a granted patent.

Data curation followed a standardized, template-driven workflow. For each record, curators extracted the target protein and its identifier, the peptide sequence and any modifications, the measurement type and value with original and standardized units, the assay/technique, and the bibliographic identifiers (DOI/PMID or patent number). Extracted fields were cross-checked and proofread within the curation team. For a given peptide ligand and protein target, multiple measurements reported under distinct assay formats or experimental conditions (for example different temperatures, buffer systems or techniques) were retained as independent records rather than averaged, in order to preserve the original experimental context and variability. Records lacking an explicitly reported peptide sequence, or in which a non-standard residue position could not be assigned a defined identity, were excluded.

On this basis, we constructed closely linked data modules. The quantitative interaction module consists of curated records of experimentally determined affinity between peptide ligands and protein targets that satisfy the above criteria. The structural module is derived from the Protein Data Bank[36] and comprises a cleaned set of peptide ligand–protein complexes in which the peptide chain contains no more than 100 residues; entries lacking a clearly defined peptide chain or a complete peptide sequence were removed. For each complex, peptide and protein chain identifiers were standardised and explicitly annotated, and we extracted peptide and protein sequences, crystallographic resolution and other basic structural descriptors, as well as residue-level interaction patterns identified by PLIP[37]. For peptide ligand–protein pairs for which both quantitative measurements and experimentally determined structures are available, the corresponding records are cross referenced through shared PDB accession codes and UniProt[38] accessions. For both modules, each record was further enriched with key physicochemical properties of the peptide ligand computed using bioinformatics tools such as RDKit and peptidy, and identifiers were

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harmonised across the resource[39,40]. A schematic overview of the construction and curation workflow is shown in Figure 1.

Figure 1. Construction and curation workflow of the PPIKB datasets.

The schematic summarises data collection from peer-reviewed articles, public patents and PDB records, multistage curation and annotation, integration of the quantitative interaction and structural modules, and cross-linking through shared identifiers.

Data Records

The curated datasets are deposited on figshare and released under a CC BY 4.0 license[50], and are also accessible through the PPIKB platform (https://ppikb.duanlab.ac/). The resource is organised into downloadable flat tables, a quantitative interaction dataset and a structural dataset, provided in delimited text (CSV) format, in which each row corresponds to a single record. The datasets can be joined on the shared UniProt and PDB identifiers to retrieve matched structure–measurement records.

Quantitative interaction dataset. Each record combines (i) target-protein information: name, sequence, organism, functional classification, subcellular localisation, gene names and UniProt accession; (ii) peptide-ligand information: name, sequence, length, SMILES where available, chemical modifications, cyclisation strategy, N- and C-terminal status, and amino-acid composition; (iii) computed physicochemical descriptors: molecular weight, aliphatic index, aromaticity, average number of rotatable bonds, net charge at pH 7, isoelectric point, hydrogen-bond donor/acceptor counts, topological polar surface area and an XlogP-like descriptor; (iv) interaction information: numeric affinity value with original and standardized units, measurement type (Kd, Ki, IC50, etc.), assay/technique, and matched PDB identifier when available; and (v) source identifiers: PMID and DOI for articles, or the patent number for patents. An example of the information captured per record is shown in Figure 2.

In the quantitative interaction dataset, the symbol X in a peptide sequence is reserved strictly for residues whose identity is undetermined. Residues with an official single-letter code are represented with that code (selenocysteine as U); non-standard or modified residues that lack an assigned single-letter code retain their position in the sequence and their identity is stated explicitly in the Chemical_Modification field (for example, “X4=N-methyltryptophan; X8=N-methylglycine”), with three-letter / PDB chemical-component identifiers where available.

Figure 2. Example of a quantitative interaction record in PPIKB.

Each affinity record is organised into five groups of fields. (A) Target protein information summarises protein name, sequence, organism, functional class, subcellular localisation, gene names and UniProt accession, together with an associated protein–protein interaction network. (B) Peptide basic information provides peptide name, sequence, length, SMILES where available, and annotation of chemical modifications, cyclisation and terminal status, together with amino acid composition. (C) Interaction information records the quantitative binding value and assay type, and links to matched structural evidence when available. (D) Peptide physicochemical descriptors are listed to support developability and property-oriented filtering. (E) Reference information links each record to the original publication or patent.

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Structural dataset. Each record summarises a curated peptide–protein complex derived from the Protein Data Bank and provides: protein chain identifier and sequence; peptide chain identifier, sequence, length and a flag for the presence of non-standard residues; PDB accession, experimental method and resolution; residue-level noncovalent contacts between the peptide and its protein partner computed using PLIP[37]; and the corresponding reference (title, year, PMID and DOI where available; the PDB accession serves as the permanent source identifier for every structural record). In the structural dataset, the Peptide_Sequence is the one-letter sequence taken directly from the corresponding PDB entry; positions occupied by non-standard or modified residues are flagged in Residue_Modification and shown as X, and the exact chemical identity of each such residue is defined by its chemical-component (HET) identifier in the source PDB entry, which is linked, displayed and available for download for every structural record. A representative record is illustrated in Figure 3.

Figure 3. Example of a structural record in PPIKB.

Each structural record describes an experimentally resolved protein–peptide complex and its residue-level interface. (A) Protein chain information lists chain identifiers and sequences. (B) Interaction information reports the PDB accession, experimental method and resolution, together with a PLIP-derived protein–peptide residue interaction diagram and labelled interaction categories. (C) Peptide chain information includes sequence, length and annotation of non-standard residues. (D) Reference information connects the structural record to the primary literature.

Data Overview

In the current release, the quantitative interaction module contains 21,845 experimentally determined peptide ligand–protein binding measurements, curated from 3,894 peer-reviewed research articles and 553 granted patents published between 1972 and 2025; the records cover 2,151 protein-target entries. The most frequent measurement types are Kd (10,607 records), IC50 (9,122) and Ki (1,714), with additional types such as EC50 (352) also represented.

Human proteins account for the majority of records (66.1%), followed by other organisms (11.5%), viruses (7.3%), Mus musculus (6.2%), Rattus norvegicus (4.9%) and bacteria (3.6%) (Figure 4A). The top human targets span multiple functional classes (Figure 4B). To enable comparison across assay readouts, affinity values were converted to a unified pAffinity scale, and the distributions for Kd, Ki and IC50 are summarised in Figure 4C.

The structural module comprises 18,005 experimentally resolved peptide–protein complexes. Structure-determination methods are dominated by X-ray diffraction (89.83%), followed by solution NMR (5.25%) and electron microscopy (4.83%); other methods are rare (Figure 4D). X-ray structures are typically determined at higher resolution than electron-microscopy structures (median 2.1 versus 3.4 Å) (Figure 4E). Residue-level interface features are summarised as a heat map of amino-acid pairing frequencies across noncovalent contacts derived from PLIP (Figure 4F), in which hydrophobic packing and acidic–basic salt-bridge pairings are prominent.

Figure 4. Data summary and statistics of the quantitative interaction and structural modules.

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(A) Taxonomic distribution of protein targets. (B) Top human protein targets by number of measurements, coloured by functional class. (C) Distributions of standardised binding strengths (pAffinity) for Kd, Ki and IC50. (D) Experimental methods for the structural complexes. (E) Resolution distributions of X-ray and electron-microscopy structures. (F) Global amino-acid pairing frequency matrix derived from noncovalent contacts (standard residues only; deduplicated by peptide–receptor sequence and best resolution).

Technical Validation

All records underwent multi-stage manual screening and quality control. Quantitative interaction records were included only when the peptide sequence was explicitly reported (no more than 100 residues), the reported value was numerical and derived from wet-lab binding measurements, and the protein target could be assigned a clearly defined sequence. Each record is linked to its originating peer-reviewed article or granted patent, enabling record-level provenance and independent verification. Protein targets and peptide ligands were mapped to standardized identifiers, and quantitative values were stored with measurement type, original and standardized units, and assay/technique descriptors. Automated checks were applied to required-field completeness, identifier validity, unit/scale normalization, and outlier flagging for manual review.

As part of release quality control, records lacking an explicitly reported peptide sequence, or containing a non-standard residue position that could not be assigned a defined identity, were corrected where the identity could be recovered from the source and otherwise removed. A small number of internally inconsistent annotations were also corrected, including topology labels contradicted by explicit cyclization information and position-indexing errors in modified-peptide entries.

For the structural module, experimentally resolved complexes were assembled from the Protein Data Bank and processed to standardize chain assignments and core metadata; residue-level noncovalent contact profiles were generated using PLIP and released alongside the structure records. The quantitative and structural modules are cross-linked via shared UniProt and PDB identifiers.

Database maintenance and updates

PPIKB follows a long-term, versioned update strategy. An automated screening pipeline is run approximately every three months to identify newly available peptide–protein interaction data, followed by manual verification and integration; each release is versioned and documented in a public update log (https://ppikb.duanlab.ac/update.php). A submission/feedback page allows the community to report issues or suggest corrections, which are reviewed and incorporated into subsequent releases. Planned content expansions include structured per-measurement assay-condition fields (buffer, pH, temperature), explicit per-residue identity fields for non-standard residues in the structural module, and additional quantitative data types (e.g., MHC–peptide and domain–peptide measurements) under a harmonized standardization scheme. We also plan to incorporate records from established external resources, in particular the structure-based protein–peptide affinity set of Wang et al. (PpISADS)[47], after re-validating each entry against its primary literature under the curation rules applied throughout PPIKB.

Usage Notes

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The curated datasets can be downloaded from the PPIKB download page and from the archived figshare snapshot[50]. Each row is a single record, and the quantitative interaction dataset and the structural dataset can be joined on the shared UniProt and PDB identifiers to retrieve matched structure–measurement pairs. Typical downstream uses include constructing training and benchmark sets for peptide–protein affinity prediction, structure-based peptide docking and scoring, sequence- and structure-based similarity retrieval, and comparative analysis of peptide–protein interfaces. The PPIKB web platform (https://ppikb.duanlab.ac/) additionally provides interactive search and visualization over the same data. Usage guidance for the search modules and analysis tools is provided on the platform's help page (https://ppikb.duanlab.ac/static/help.php).

PPIKB currently focuses on direct quantitative binding constants (Kd, Ki, IC50 and related measures) between peptide ligands and defined protein targets curated from primary literature and patents. Specialized data types such as MHC–peptide binding measurements, which are extensively maintained by dedicated resources (e.g., IEDB), and semi-quantitative domain–peptide interactome readouts differ in measurement nature and require a separate standardization scheme; their systematic, harmonized incorporation is planned for future versions.

Data availability

The datasets generated and analysed in this study are available from the PPIKB platform (https://ppikb.duanlab.ac/) and are archived on figshare (https://doi.org/10.6084/m9.figshare.31082407), released under a CC BY 4.0 license.

Code availability

Custom scripts used for data processing, curation and quality control are available from the corresponding author upon reasonable request. The PPIKB web platform will be kept operating and freely accessible for the foreseeable future (https://ppikb.duanlab.ac/).

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Author contributions: N.Z. designed and built the PPIKB platform and drafted the manuscript. Y.M. performed data cleaning and standardization. C.Z. verified and proofread the curated data. S.C., C.L., Y.W. and J.G. collected and curated subsets of the data. H.D. conceived and supervised the project. All authors reviewed and approved the final manuscript.

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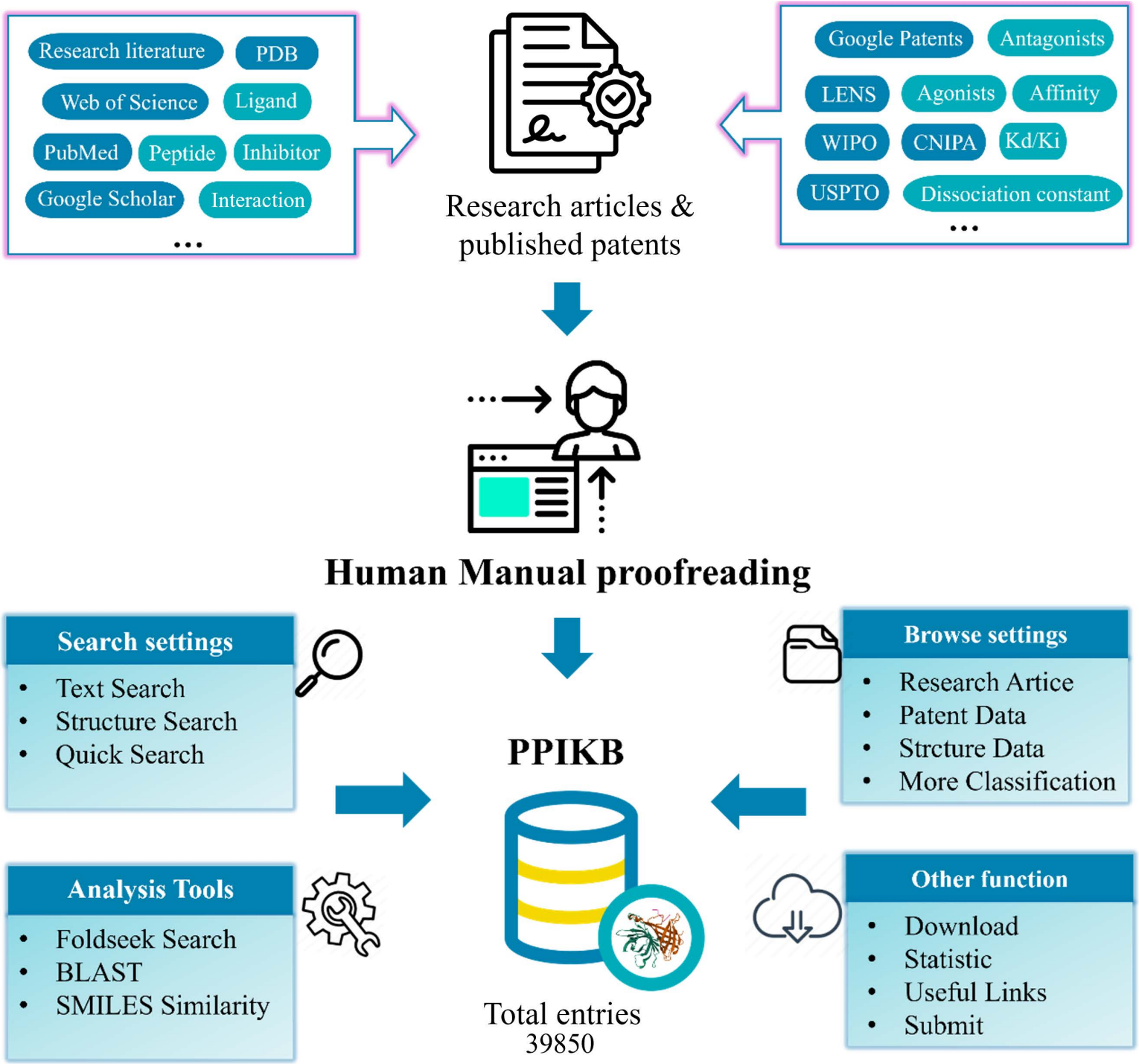
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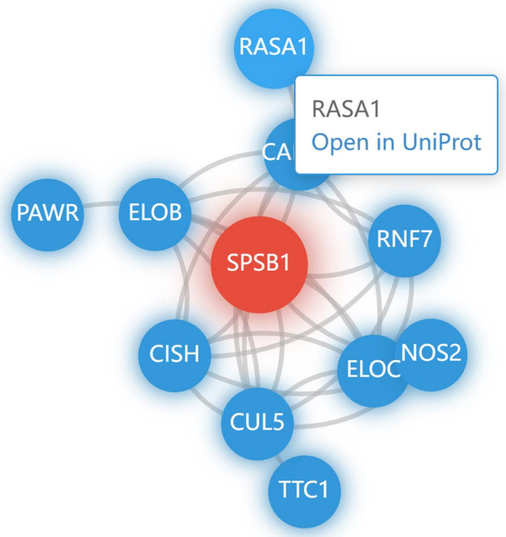

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Total entries
39850

A

Target Protein Information

Protein_Name	SPRY domain-containing SOCS box protein 1
Protein_Sequence	MGQKVTGGIKTVDMRDPTYRPLKQELQGLDYCKPTRLDLLLDMPPVSYD VQLLHSWNNNDRSLNVFVKEDDKLIFHRHPVAQSTDAIRGKVGYTRGLH VWQITWAMRQRGTHAVVGVATADAPLHSVGYTTLVGNNHESWGWDLGR NRLYHDGKNQPSKTYPAFLEPDETFIVPDSFLVALDMDDGTLSFIVDGQY MGVAFRGLKGKKLYPVVSAVWGHCEIRMRYLNGLDPEPLPLMDLCRRSV RLALGRERLGEIHTLPLPASLKAYLLYQ
Organism_Source	Homo sapiens
Functional_Classification	E3 ubiquitin ligases
Cellular_Localization	Cytoplasm
Gene_Names	SPSB1
UniProt_ID	Q96BD6

Protein-Protein Interaction Networks

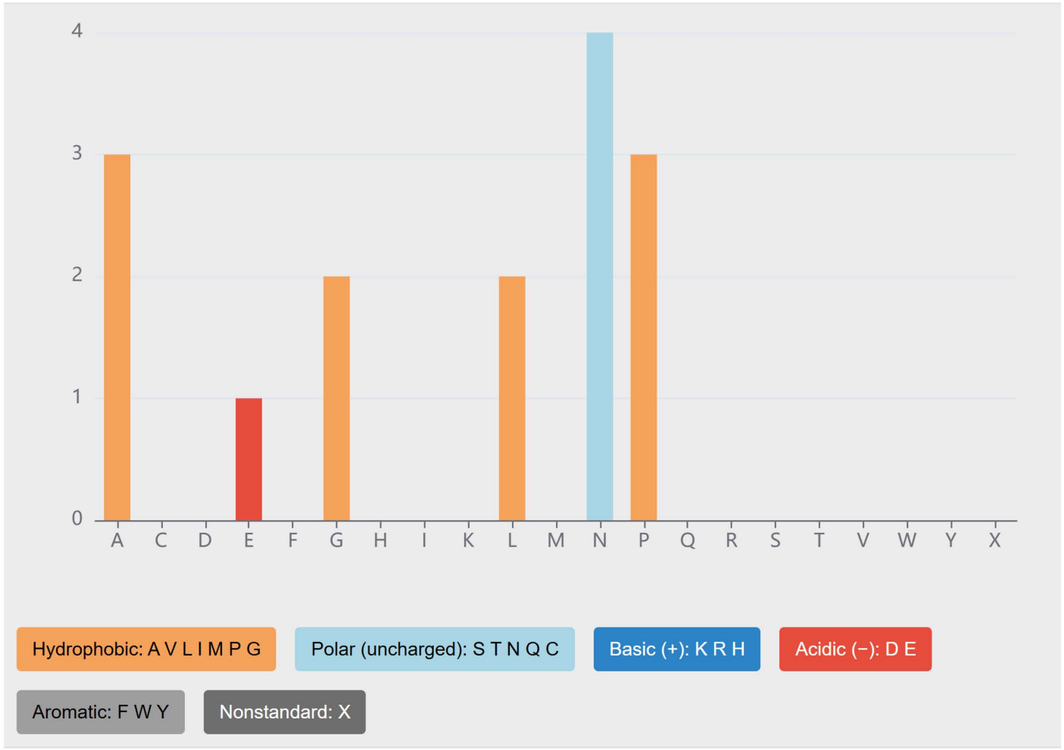


B

Peptide Basic Information

Peptide_Name	Par-4(67-81)
Peptide_Sequence	NELNNNLPGGAPAAP
Peptide_Length	15
Peptide_SMILES	CC(C)C[C@H](NC(=O)[C@H](CCC(=O)O)NC(=O)[C@@H](N)CC(N)=O)C(=O)N[C@@H](CC(N)=O)C(=O)N[C@@H](CC(N)=O)C(=O)N[C@@H](CC(N)=O)C(=O)N[C@@H](CC(C)C)C(=O)N1CCC[C@H]1C(=O)NCC(=O)NCC(=O)N[C@@H](C)C(=O)N1CCC[C@H]1C(=O)N[C@@H](C)C(=O)N[C@@H](C)C(=O)N1CCC[C@H]1C(=O)O
Chemical_Modification	None
Cyclization_Method	None
Linear/Cyclic	Linear
N-terminal_Modification	Free
C-terminal_Modification	Free

Amino_Acid_Distribution

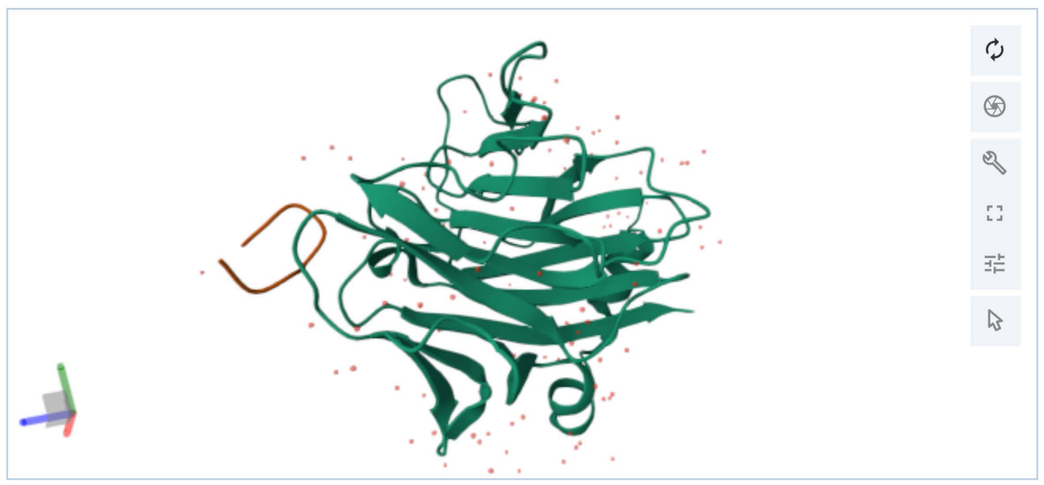


C

Interaction Information

Affinity	KD=150 nM
Affinity_Assay	Isothermal titration calorimetry
PDB_ID	2JK9
Type	Affinity ligand

Structure



D

Peptide Physicochemical

Molecular_Weight	1448.56	**Isoelectric_Point**	3.84998
Aliphatic_Index	72.00000	**Hydrogen_Bond_Acceptors**	21
Aromaticity	0.00000	**Hydrogen_Bond_Donors**	18
Average_Rotatable_Bonds	2.73333	**Topological_Polar_Surface_Area**	654.01000
Charge_at_pH_7	-1.00024	**X_logP_energy**	-9.51480

E

Reference Information

Document_Type	Research Articles
Title	Structural basis for Par-4 recognition by the SPRY domain- and SOCS box-containing proteins SPSB1, SPSB2, and SPSB4.
Release_Year	2010
PMID	20561531
DOI	10.1016/j.jmb.2010.06.017

A

Protein Information

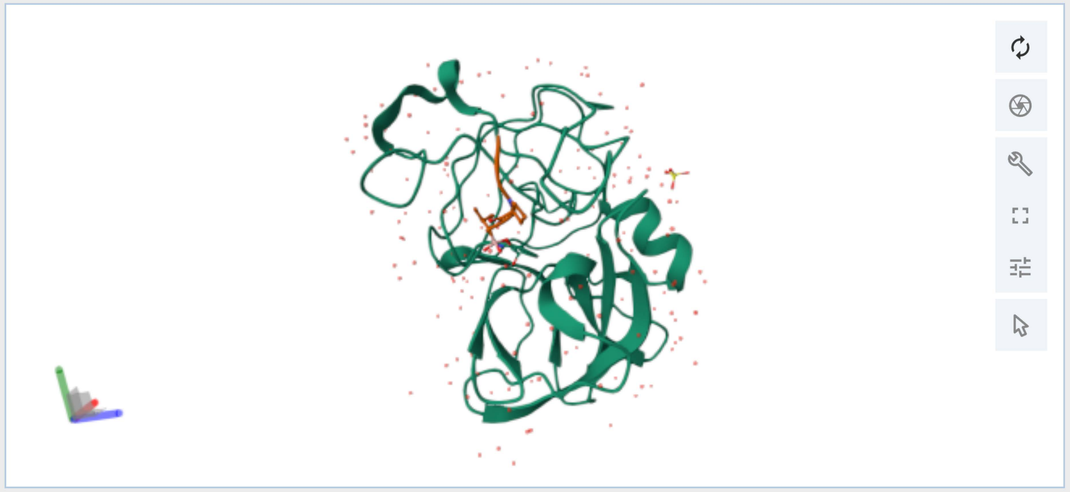
Protein_Chain	A
Protein_Sequence	ANIVGGIEYSINNASLCSVGFSVTRGATKGFVTAGHCGTVNATARIGGAVV GTFAARVFPGNDRAWVSLTSAQTLLPRVANGSSFVTVRGSTEAAVGAAV CRSGRTTGYQCGTITAKNVTANYAEGAVRGLTQGNACAGRGDSGGSWIT SAGQAQGVMSGGNVQSNGNNCGIPASQRSSLFERLQPILSQYGLSLVTG

B

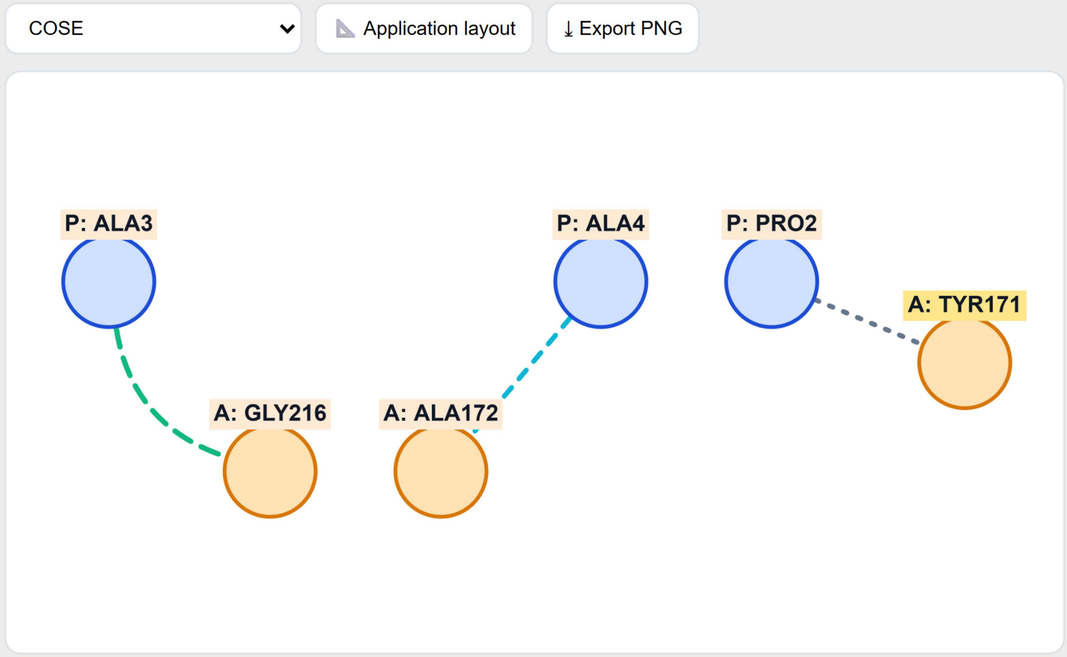
Interaction Information

PDB_ID	2LPR
Method	X-RAY DIFFRACTION
Resolution	2.25 angstrom

Structure



Protein-peptide residue interaction



Interaction_xlsx	Download interaction table (.xlsx)

C

Peptide Information

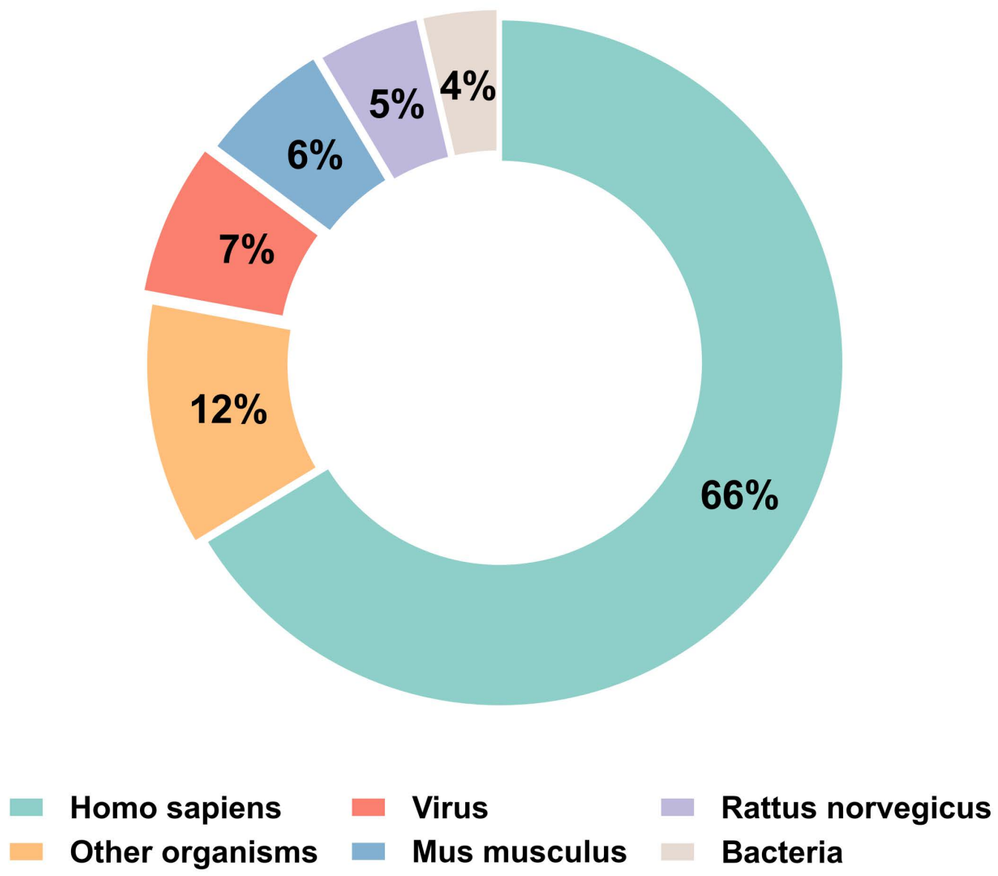
Peptide_Chain	B[auth P]
Peptide_Sequence	XAAPV
Peptide_Length	5
Non-standard residues	Yes

D

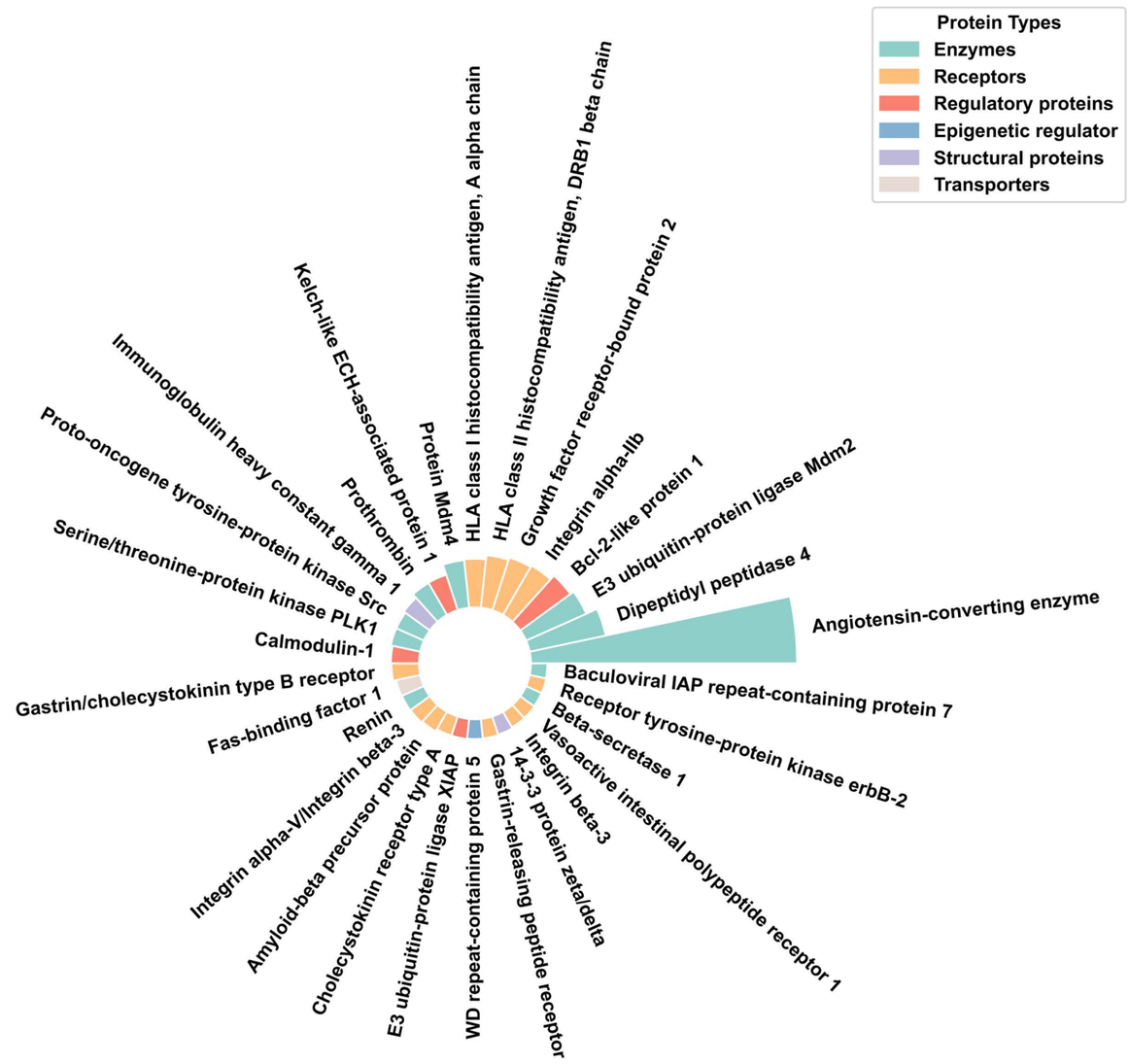
Reference Information

Title	Structural basis for broad specificity in alpha-lytic protease mutants.
Release_Year	1991
PMID	1931963
DOI	10.1021/bi00107a005

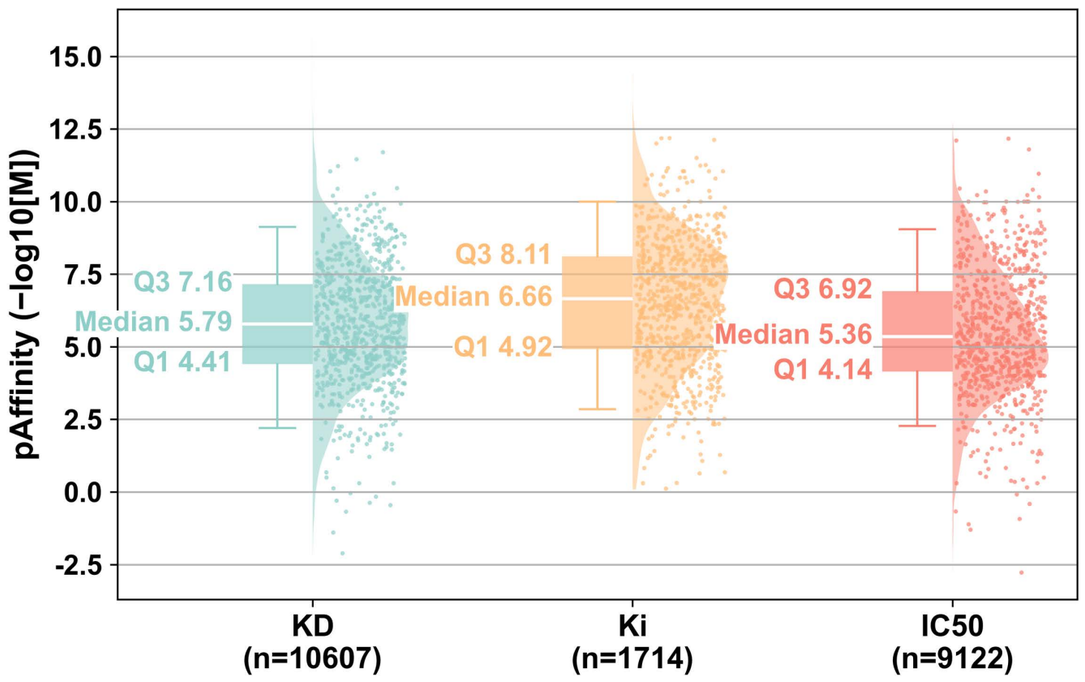
A


66%
12%
7%
6%
5%
4%
Homo sapiens
Other organisms
Virus
Mus musculus
Rattus norvegicus
Bacteria

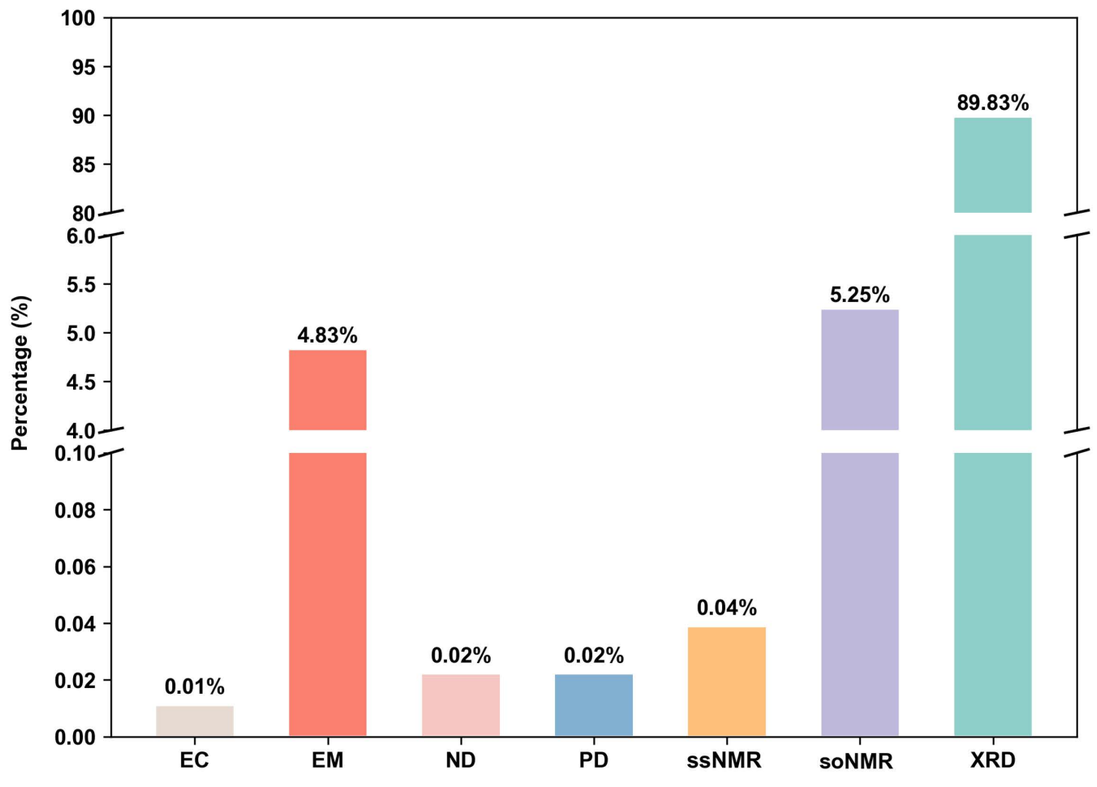
B


Protein Types
Enzymes
Receptors
Regulatory proteins
Epigenetic regulator
Structural proteins
Transporters
Kelch-like ECH-associated protein 1
Immunoglobulin heavy constant gamma 1
Prothrombin
Protein Mdm4
HLA class I histocompatibility antigen, A alpha chain
HLA class II histocompatibility antigen, DRB1 beta chain
Growth factor receptor-bound protein 2
Integrin alpha-IIb
Bcl-2-like protein 1
E3 ubiquitin-protein ligase Mdm2
Dipeptidyl peptidase 4
Angiotensin-converting enzyme
Baculoviral IAP repeat-containing protein 7
Receptor tyrosine-protein kinase erbB-2
Beta-secretase 1
Vasoactive intestinal polypeptide receptor 1
Integrin beta-3
14-3-3 protein zeta/delta
Gastrin-releasing peptide receptor
WD repeat-containing protein 5
E3 ubiquitin-protein ligase XIAP
Cholecystokinin receptor type A
Amyloid-beta precursor protein
Integrin alpha-V/Integrin beta-3
Renin
Fas-binding factor 1
Gastrin/cholecystokinin type B receptor
Calmodulin-1
Serine/threonine-protein kinase PLK1
Proto-oncogene tyrosine-protein kinase Src

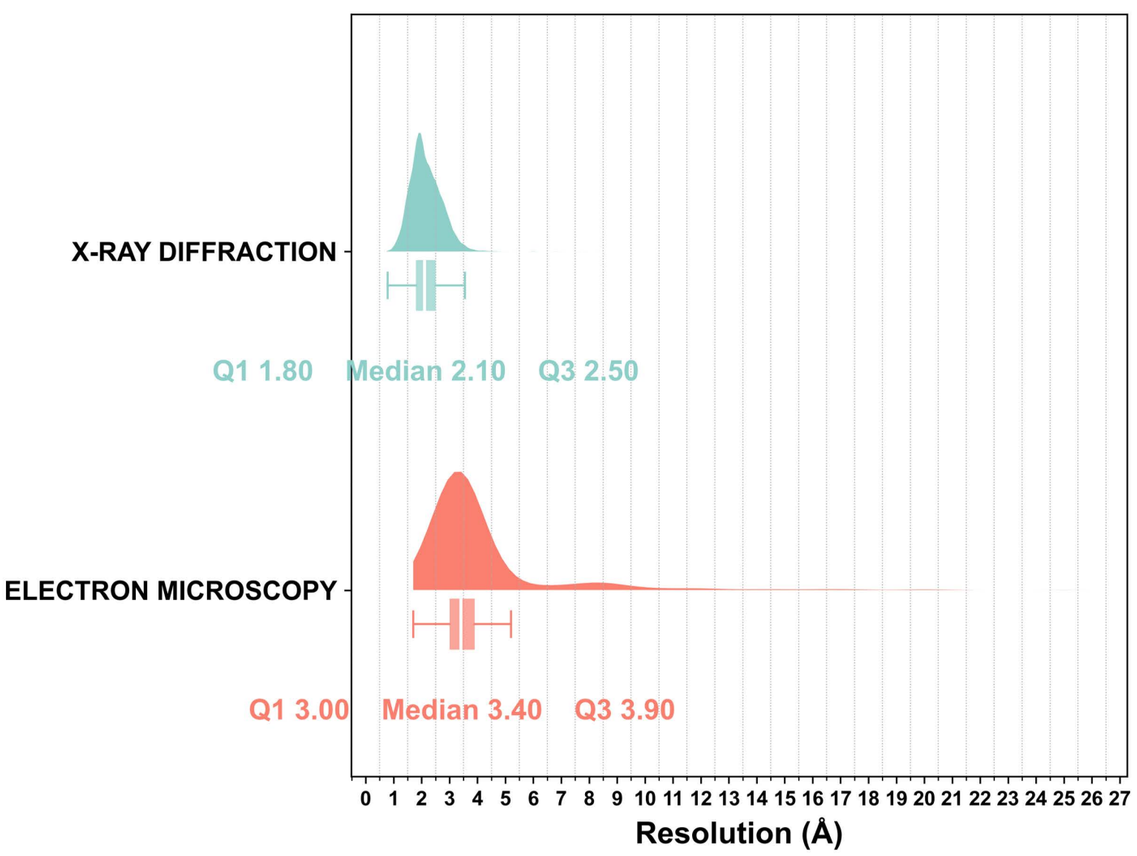
C


pAffinity (−log10[M])
15.0
12.5
10.0
7.5
5.0
2.5
0.0
-2.5
Q3 7.16
Median 5.79
Q1 4.41
Q3 8.11
Median 6.66
Q1 4.92
Q3 6.92
Median 5.36
Q1 4.14
KD
(n=10607)
Ki
(n=1714)
IC50
(n=9122)

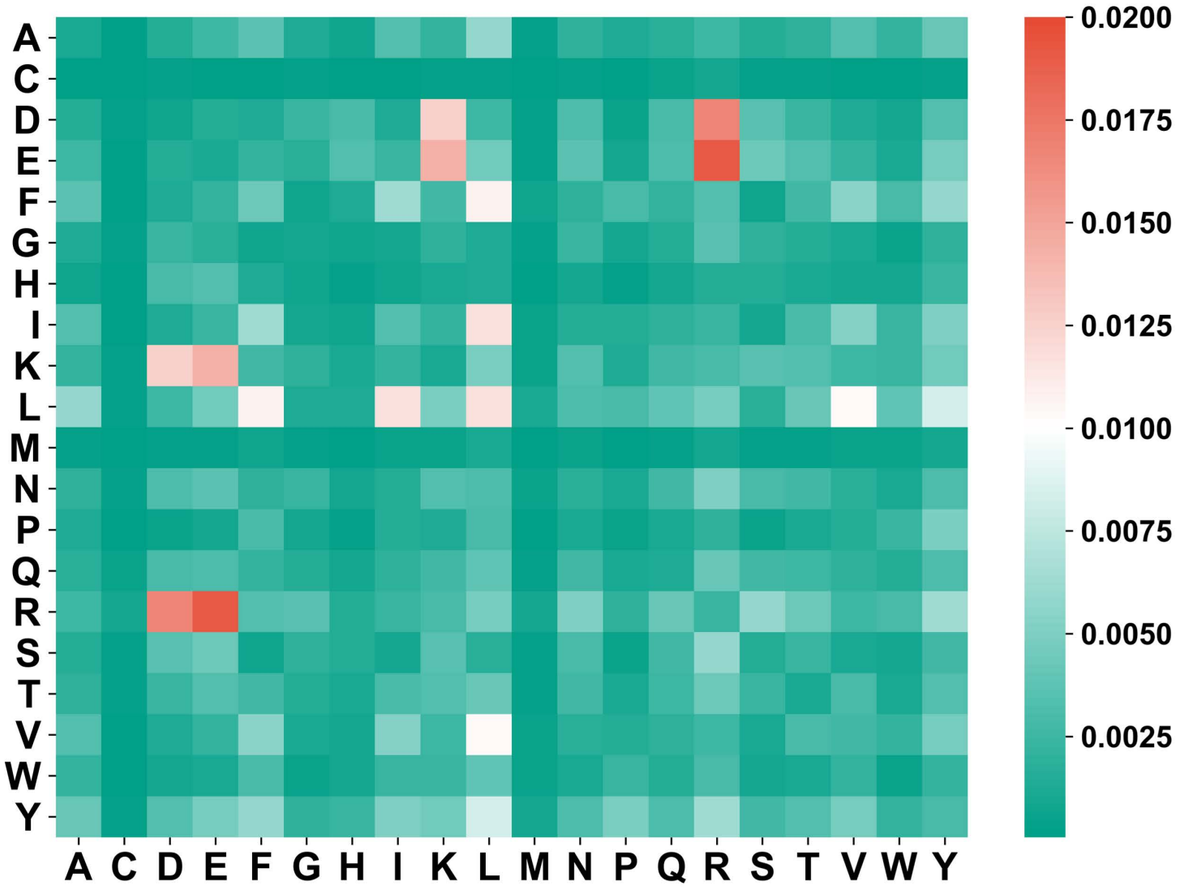
D


Percentage (%)
100
95
90
85
80
6.0
5.5
5.0
4.5
4.0
0.10
0.08
0.06
0.04
0.02
0.00
0.01%
4.83%
0.02%
0.02%
0.04%
5.25%
89.83%
EC
EM
ND
PD
ssNMR
soNMR
XRD

E


X-RAY DIFFRACTION
Q1 1.80
Median 2.10
Q3 2.50
ELECTRON MICROSCOPY
Q1 3.00
Median 3.40
Q3 3.90
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27
Resolution (Å)

F


A
C
D
E
F
G
H
I
K
L
M
N
P
Q
R
S
T
V
W
Y
0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025